12 - Relations

Week 13 – 11 Nov 2025

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Cheat sheet

Definition

Definition

Let R be a relation on a set A.

1. R is **reflexive** if and only if for all $x \in A$, $x\ R\ x$.
2. R is **symmetric** if and only if for all $x, y \in A$, if $x\ R\ y$, then $y\ R\ x$.
3. R is **transitive** if and only if for all $x, y, z \in A$, if $x\ R\ y$ and $y\ R\ z$, then $x\ R\ z$.

1. *R is reflexive* $\iff$ *for all x in A, $(x, x) \in R$.*
2. *R is symmetric* $\iff$ *for all x and y in A, if $(x, y) \in R$ then $(y, x) \in R$.*
3. *R is transitive* $\iff$ *for all x, y, and z in A, if $(x, y) \in R$ and $(y, z) \in R$, then $(x, z) \in R$.*

Definition of an Equivalence Relation, Equivalence Class

Definition

Let A be a set and R a relation on A.

R is an **equivalence relation** if and only if R is reflexive, symmetric, and transitive.

Definition

Suppose A is a set and R is an equivalence relation on A. For each element a in A, the **equivalence class of a**, denoted by $[a]$, is the set of all elements $x \in A$ such that x is related to a by R.

$$[a] = \{x \in A | x\ R\ a\}.$$

$$\text{for all } x \in A, \quad x \in [a] \iff x\ R\ a.$$

Ex. 8.1.11. Let $A = \{3, 4, 5\}$ and $B = \{4, 5, 6\}$ and let S be the "divides" relation. That is,

$$\text{for all } (x, y) \in A \times B, \quad x \; S \; y \quad \iff \quad x | y.$$

State explicitly which ordered pairs are in S and S^{-1}?

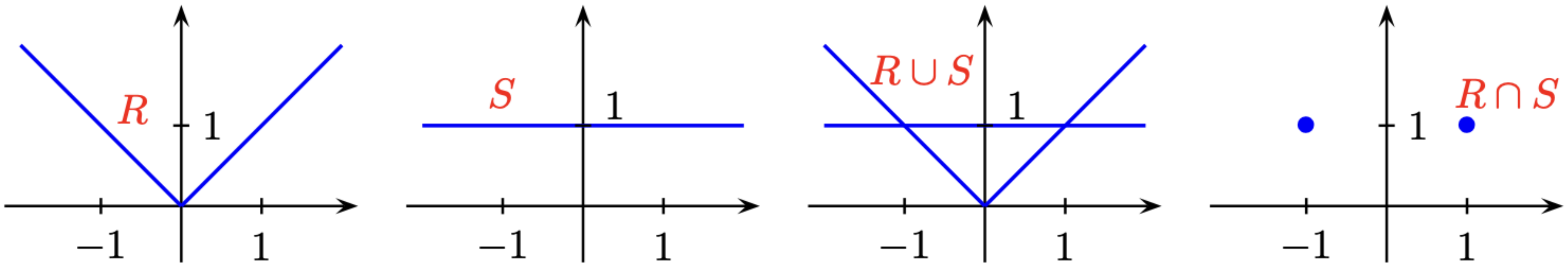
Ex. 8.1.14. Define a relation S on $B = \{a, b, c, d\}$ by $S = \{(a, b), (a, c), (b, c), (d, d)\}$. Draw the directed graphs of the relation S.

Ex. 8.1.22. Define relations R and S from $\mathbb{R}$ to $\mathbb{R}$ as follows:

$$R = \{(x, y) \in \mathbb{R} \times \mathbb{R} |\ y = |x|\} \quad \text{and}$$
$$S = \{(x, y) \in \mathbb{R} \times \mathbb{R} |\ y = 1\}.$$

Graph R, S, $R \cup S$, and $R \cap S$ in the Cartesian plane.

SOLUTION .



□

Ex. 8.2.17, 21. Determine whether the given relation is reflexive, symmetric, transitive, or none of these. Justify your answers.

17. Recall that a prime number is an integer that is greater than 1 and has no positive integer divisors other than 1 and itself. (In particular, 1 is not prime.) A relation P is defined on $\mathbb{Z}$ as follows:

$$\text{For all } m, n \in \mathbb{Z},\ m\ P\ n \quad \Longleftrightarrow \quad \exists \text{ a prime number } p \text{ such that } p|m \text{ and } p|n.$$

SOLUTION . **17.**

a. <u>P is not reflexive.</u>

Consider the integer 1. By Theorem 4.3.2, we know that the only divisors of 1 are 1 and -1. Since prime numbers are greater than 1, no prime numbers are divisors of 1. Therefore $\nexists$ prime number p such that $p|1$ and $p|1$. Therefore $1 \not{P} 1$.

b. <u>P is symmetric.</u>

Suppose m and n are integers such that $m\ P\ n$. Then for some prime number p, we have $p|m$ and $p|n$.

Therefore, for the same prime number p, $p|n$ and $p|m$. Therefore, $n\ P\ m$.

c. <u>P is not transitive.</u> Consider the integers $2, 6, 9$. Then 2 is a prime number and $2|2$ and $2|6$, hence $2\ P\ 6$. Also, 3 is a prime number and $3|6$ and $3|9$, hence $6\ P\ 9$. However, $2 \not{P} 9$ since 2 and 9 do not have any common prime factor.

21. Let $X = \{a, b, c\}$ and $\mathscr{P}(X)$ be the power set of X. A relation $\mathbf{L}$ is defined on $\mathscr{P}(X)$ as follows:

$$\text{For all } A, B \in \mathscr{P}(X),\ A \mathbf{L} B \quad \Longleftrightarrow \quad |A| < |B|.$$

($|A| < |B|$ means the number of elements in A is less than the number of elements in B.)

21.

a. <u>$\mathbf{L}$ is not reflexive.</u>

Consider the set $A = \{a\}$ which is a subset of X. Then $|A| = 1$ and $1 \nless 1$. Therefore, $|A| \nless |A|$. Therefore $(A, A) \notin \mathbf{L}$.

b. <u>$\mathbf{L}$ is not symmetric.</u>

Consider the subsets $A = \varnothing$ and $B = \{a\}$ of X. Then $|A| = 0$ and $|B| = 1$. Since $0 < 1$, we have $(A, B) \in \mathbf{L}$. Since $1 \nless 0$, we know that $(B, A) \notin \mathbf{L}$.

c. <u>$\mathbf{L}$ is transitive.</u>

Suppose A, B and C are subsets of X such that $A \mathbf{L} B$ and $B \mathbf{L} C$. Then we know that $|A| < |B|$ and $|B| < |C|$. By the transitivity of $<$, we have $|A| < |C|$. Hence $A \mathbf{L} C$.

$\square$

Ex. 8.2.40–42. Suppose R and S are relations on a set A.

40. If R and S are reflexive, is $R \cup S$ reflexive? Why?

41. If R and S are symmetric, is $R \cup S$ symmetric? Why?

42. If R and S are transitive, is $R \cup S$ transitive? Why?

SOLUTION . **40.** Yes. Let a be an element in A. Since R is reflexive, we know that $(a, a) \in R$. Hence, by the definition of $\cup$, we have $(a, a) \in R \cup S$.

41. Yes. Suppose we are given $a, b \in A$ such that $(a, b) \in R \cup S$. Then, either $(a, b) \in R$ or $(a, b) \in S$.

Case 1: $(a, b) \in R$. Since R is symmetric, we have $(b, a) \in R$. Hence, by definition of $\cup$, $(b, a) \in R \cup S$.

Case 2: $(a, b) \in S$. Since S is symmetric, we have $(b, a) \in S$. Hence, by definition of $\cup$, $(b, a) \in R \cup S$.

42. No, $R \cup S$ need not be transitive.

Counterexample:

Consider $A = \{a, b, c\}$, and $R = \{(a, b)\}$ and $S = \{(b, c)\}$. Then $R \cup S = \{(a, b), (b, c)\}$. Clearly R and S are transitive (true by default since there is nothing to check).

We note that $(a, b) \in R \cup S$ and $(b, c) \in R \cup S$ but $(a, c) \notin R \cup S$. $\square$

Ex. 8.3.15. Determine which of the following congruence relations are true and which are false.

a. $17 \equiv 2 \pmod 5$ b. $4 \equiv -5 \pmod 7$

c. $-2 \equiv -8 \pmod 3$ d. $-6 \equiv 22 \pmod 2$.

$$m \equiv n \pmod d \iff d|(m-n).$$

SOLUTION . a. True since $17 - 2 = 15 = 5 \cdot 3$ and hence $5|15$.
b. False since $4 - (-5) = 9$ and $7 \nmid 9$.
c. True since $-2 - (-8) = 6 = 3 \cdot 2$ and hence $3|6$.
d. True since $-6 - 22 = -28 = 2 \cdot (-14)$ and hence $2|(-28)$.

Ex. 8.3.32. Let A be the set of all straight lines in the Cartesian plane. Define a relation $||$ on A as follows:

$$\text{For all } l_1 \text{ and } l_2 \text{ in } A, \quad l_1 \mathrel{||} l_2 \quad \iff \quad l_1 \text{ is parallel to } l_2.$$

Prove that $||$ is an equivalence relation. Describe the distinct equivalence classes of the relation.

Ex. 8.3.46. Let R be a relation on a set A and suppose R is a symmetric and transitive relation. Prove the following:

If for every x in A there is a y in A such that $x \; R \; y$, then R is an equivalence relation.

Ex. 8.3.33. Let A be the set of points in the rectangle with x and y coordinates between 0 and 1. That is,

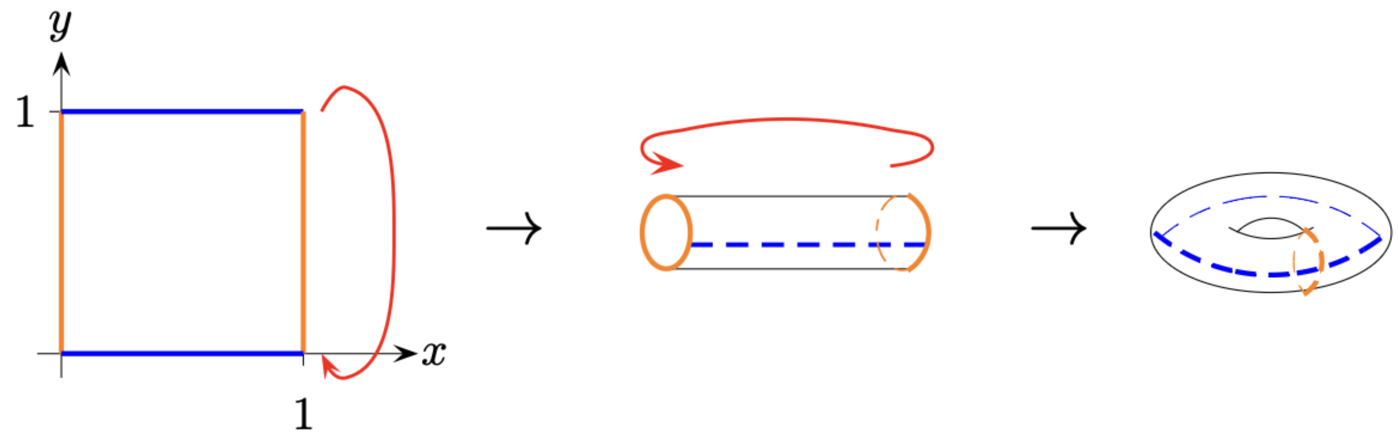
$$A = \{(x, y) \in \mathbb{R} \times \mathbb{R} | \ 0 \leq x \leq 1 \quad \text{and} \quad 0 \leq y \leq 1\}.$$

Define a relation R on A as follows: For all (x_1, y_1) and (x_2, y_2) in A,

$$\begin{aligned}
(x_1, y_1) \ R \ (x_2, y_2) \quad \iff \quad & (x_1, y_1) = (x_2, y_2); \ \text{or} \\
& x_1 = 0 \ \text{ and } \ x_2 = 1 \ \text{ and } \ y_1 = y_2; \ \text{ or} \\
& x_1 = 1 \ \text{ and } \ x_2 = 0 \ \text{ and } \ y_1 = y_2; \ \text{ or} \\
& y_1 = 0 \ \text{ and } \ y_2 = 1 \ \text{ and } \ x_1 = x_2; \ \text{ or} \\
& y_1 = 1 \ \text{ and } \ y_2 = 0 \ \text{ and } \ x_1 = x_2; \ \text{ or} \\
& (x_1, y_1), (x_2, y_2) \in \{(0,0), (0,1), (1,0), (1,1)\}.
\end{aligned}$$

In other words, all points along the top edge of the rectangle are related to the points along the bottom edge directly beneath them, and all points directly opposite each other along the left and right edges are related to each other. The points in the interior of the rectangle are not related to anything other than themselves. Then R is an equivalence relation on A. Imagine gluing together all the points that are in the same equivalence class. Describe the resulting figure.

SOLUTION .



The distinct equivalence classes can be identified with the points on a geometric figure, called a torus, that has the shape of the surface of a doughnut (or donut). Each point in the interior of the rectangle $\{(x, y)|\ 0 < x < 1 \text{ and } 0 < y < 1\}$ is only equivalent to itself. Each point on the top edge of the rectangle is in the same equivalence class as the point vertically below it on the bottom edge of the rectangle (so we can imagine identifying these points by gluing them together – this gives us a cylinder), and each point on the left edge of the rectangle is in the same equivalence class as the point horizontally across from it on the right edge of the rectangle (so we can also imagine identifying these points by gluing them together – this brings the two ends of the cylinder together to produce a torus).

Bonus Qn: AY21/22 Finals

QUESTION 7. (16 marks)

(a) For the relation R below defined on the set $\mathbb{R}^2$, determine if it is reflexive, if it is symmetric and if it is transitive. Justify your answer.

$$(x_1, x_2)\ R\ (x_3, x_4) \leftrightarrow x_i = x_j \text{ for some } i \neq j, \text{ and } i, j = 1, 2, 3 \text{ or } 4.$$

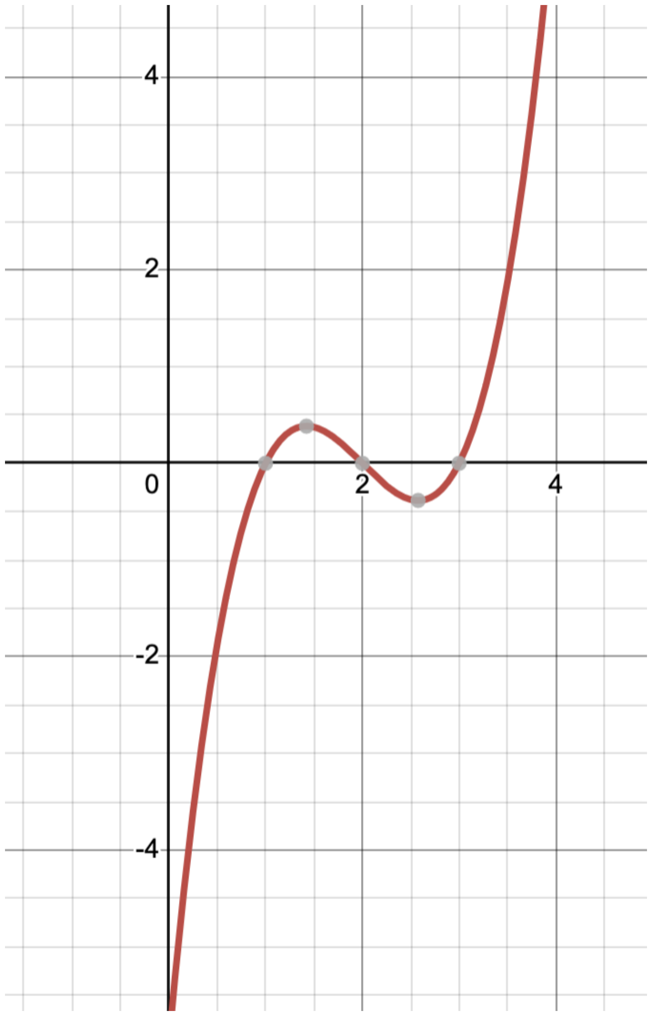
(b) Supose that T is a reflexive relation on a set A such that for every $x, y, z \in A$, if $x\ T\ y$ and $x\ T\ z$ then $y = z$. Show that T is an equivalence relation, and describe the equivalence classes of T.

Finals Prediction

- Logical proves using Propositional Logic (Tut 2)
  - Eg. Modus Ponens etc.
- Everything after midterm will be tested
  - Relations
  - Induction (Strong and weak)
  - Set Theory
  - **Functions**
  - Algorithms (Euclidean or Division)
  - Complex numbers – Finding all the roots
- Definition questions (Copy all definitions into your cheat-sheet)

Commons functions (Real)

- Injective but not surjective: y = e^x
- Surjective but not injective: some polynomials like y = (x-1)(x-2)(x-3)
- Neither: y = x^2



Good luck and get A+